

GENERAL SPECIFICATION G3.10

Earthworks

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G3.10 EARTHWORKS

G3.10.1 Scope

- (a) This Section of the Specification contains requirements which, where relevant to this Contract shall apply to site clearance; excavation including trench excavation; embankments; filling; backfilling; top soiling and turfing.
- (b) Excavation requiring blasting should refer to the respective section of the general specifications.
- (c) This Specification should be read in conjunction with Particular Specification on Environmental Considerations and Protection.

G3.10.2 Definitions

- (a) The following terms shall have the meanings hereby assigned to them:

"Topsoil"	Any surface material, including turf, suitable for use in soiling areas to be grassed or cultivated.
"Rock"	Material which in the opinion of the S.O. would normally have to be loosened either by blasting or using pneumatic tools (excluding clay spades powered by an air compressor) or by other rock quarrying methods or, if excavated by hand, using wedge and sledge hammers. An isolated solid boulder or detached piece of rock shall qualify as Rock only if it exceeds 0.5 m3 in volume.
"Bulk Excavation"	Excavation in open cut (excluding Trench Excavation and Incidental Excavation) down to levels specified on the Drawings or otherwise as being the general levels after completion of excavation.
"Trench Excavation"	Excavation of trenches into which pipes or cables are to be laid to levels and limits specified on the Drawings or otherwise directed by the S.O..
"Incidental Excavation"	Excavation below or outside the limits of Bulk Excavation and Trench Excavation.
"General Excavation"	Excavation, whether Bulk or Incidental Excavation, required for structures but excludes Trench Excavation.
"Excess Excavation"	Excavation outside the limits specified or shown on the Drawings or directed by the S.O. for Bulk, Trench or Incidental Excavation.
"General Fill"	Approved material whether excavated within the Site or imported from outside the Site which is used and compacted to form general embankments and general platforms above original or stripped ground levels. Excludes Engineering Fill.

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| "Engineering Fill" | Approved material whether excavated within the Site or imported from outside the Site which is used and compacted to form embankments and platforms for foundations to structures as shown on the Drawings. |
| "Backfill" | Excavated material which is refilled or backfilled and compacted in pipe trenches and in the spaces between concrete structures and the sides of the excavations. |
| "Fields" | Fields, plantations, padi, grass verges, unpaved areas of the Site and the like. |
- (b) All reference to "excavation" or "excavations" shall apply to Bulk, Trench, Incidental and Excess Excavation(s) except where clearly stated to the contrary.
- (c) All reference to "fill" shall apply to both General Fill and Engineering Fill except where clearly stated to the contrary.

G3.10.3 Record Surveys

- (a) Before the surface of any part of the Site is disturbed or the works thereon are begun the Contractor shall take and record levels of any such part. Immediately following completion of excavation or filling in any area, the Contractor shall also take and record levels of the finished surface. The levels shall be taken and recorded in the manner specified or as agreed with the S.O. in his presence and such levels when agreed with him shall form the basis for measurement.
- (b) The Contractor shall employ a licensed surveyor for taking and recording such levels, who shall certify the results as correct.
- (c) The Contractor shall obtain the S.O.'s prior approval to the proposed methods of obtaining and recording this information and shall afford the S.O. all facilities to attend and check each survey and recording. The Contractor shall give the S.O. notice of his intention to carry out such survey work at least 24 hours in advance of the commencement of the survey. The Contractor shall carry out to the S.O.'s instructions all extra surveys required to resolve any doubts which may arise as to the correctness of any survey or record and the S.O.'s decision shall be final regarding what shall be recorded as the correct survey.
- (d) Each survey drawing which has been agreed shall be referred to as a "Record Survey", shall be signed by the Contractor and the S.O. and shall form the basis of measurement of the appropriate works. The Contractor shall supply four (4) hard copies of each Record Survey to the S.O.. Two copies of each Record Survey shall be on A1 size paper, one copy of each on A1 size plastic transparency and one copy of each on A3 size paper. The hard copies shall be bound as loose-leaf volumes. In addition, the Contractor shall also provide one disk copy of each Record Survey in the same format as specified for As Built drawings on DVD-R.

G3.10.4 Disposal of Materials

- (a) All requirements herein for the disposal by the Contractor of materials arising from site clearance or from excavations are subject to the provisions of the Specification.

G3.10.5 Excess Excavation to be Made Good

- (a) The Contractor at his own expense shall remove from the Site all material resulting from Excess Excavation and shall make good the same with such kind of fill material or in such class of concrete as may be reasonably required by the S.O. having regard to the circumstances.

G3.10.6 Site Clearance

- (a) All areas of the Site marked on the Drawings or specified for clearance or from which material is to be excavated or upon which filling is to be deposited shall be cleared to the extent required by the S.O. of all buildings, walls, gates, fences and other structures and obstructions and of all bushes, hedges, trees, stumps, roots and other vegetation except for trees marked for preservation. Material so cleared shall be disposed of off the Site as directed by the S.O.. No boundary stones, survey pegs, reference markers or temporary bench marks and site investigation markers shall be disturbed from its original position by the Contractor during site clearance and should such stones/markers be disturbed as a result thereof, the contractor shall be required to resurvey and reinstate at his own cost to the satisfaction of S.O. or reimburse the Board the cost of resurvey and materials etc., which shall be deducted by way of set off from any sums due or which shall come due from the Board to the Contractor.

G3.10.7 Trees

- (a) Trees shall be uprooted or cut down as near to ground level as possible if uprooting is not feasible and disposed of off the Site by the Contractor. The burning of trees and other vegetation shall not be permitted on Site. Trees marked for preservation shall be retained in their original state.

G3.10.8 Stumps

- (a) Except where drainage blanket or other fill material is to be placed on areas of soft alluvium or swamp as directed by the S.O., stumps and roots whether existing or remaining after tree felling shall be grubbed up and disposed of off the Site. Stumps, roots and surface vegetation shall be disturbed as little as possible where drainage blanket or other fill material is to be placed on areas of soft alluvium or swamp. No stumps shall be buried in any fill or embankment.

G3.10.9 Stripping Topsoil

- (a) Where ordered by the S.O. or elsewhere in the contract documents topsoil shall be stripped, to such depths and over such areas as he may direct, as a separate operation prior to any further excavation which may be required.

- (b) The topsoil shall be transported to a location on the Site proposed by the Contractor and approved by the S.O. and stockpiled in temporary tips for use in the Works. The height of these temporary topsoil tips shall be limited to 3 metres unless otherwise permitted by the S.O.. The exposed faces of these tips shall be grass seeded to minimise erosion. The Contractor shall construct and maintain any necessary access to the tips. The cost of such tips, grass seeding and access shall be deemed to be included in the Contract Rates for stripping topsoil.

G3.10.10 General Excavation

- (a) The ground shall be excavated by such methods and to such dimensions and depths as shall allow for the proper construction of the Works.
- (b) The excavation shall proceed in such a manner that there is always a general fall in the excavated area towards the lower contours. The excavated area shall in any case, always be maintained in such a state that there are no stagnant pools of water present, especially after rain. Where stagnant pools develop, all softened material shall be excavated and the resulting void backfilled with approved material to the satisfaction of the S.O. at no additional cost.
- (c) Where nominal 'payment' limits of excavation are not shown on the Drawings or otherwise specified they shall be deemed to be the minimum net limits which would allow the outline of the completed structure to be lowered vertically from ground level into its final position. The Contractor shall make his own allowance for any working space required and any excavation outside the aforesaid limits which has not been ordered by the S.O. whether it be due to slips, falls, over excavation, or required for timbering, shuttering or working space or otherwise to suit the Contractor's method of working or unavoidable over break or due to his carelessness or error or from any other clause shall be held to be Excess Excavation.

G3.10.11 Trench Excavation

- (a) The line and level of trenches shall be as shown on the Drawings or as directed by the S.O.. Before commencing Trench Excavation, the route of the trench shall be pegged out accurately and the existing ground levels shall be agreed with the S.O.. Strong sight rails shall then be fixed and maintained at each change of gradient, and at as many intermediate points as may be necessary. On these rails shall be marked the centre line and the level to which the excavation is to be carried out, such rails being not more than 35 m apart.
- (b) Trench Excavation shall be carried out by such methods and to such lines, dimensions and depths as shall allow for the proper construction of the Works, provided always that, unless the S.O. permits otherwise and subject to any specific requirements of the Specification no Trench Excavation shall be less than the nominal clear widths stated below:
- (c) The maximum clear width of the trench shall be not greater than the minimum clear width stated below plus 150 mm unless otherwise permitted by the S.O. on Site. For steel pipelines, the clear width of the trench shall be kept to the minimum width consistent with the backfill-compaction equipment and the type of joint used to avoid imposing excessive external loads on the pipe.

External diameter of pipe including any sheathing (D)	Clear width of trench for backfill with excavated material	Clear width of trench for backfill with sand/imported granular material
Not exceeding 500 mm	D + 450 mm	D + 450 mm
Exceeding 500 mm but not exceeding 750 mm	D + 600 mm	D + 600 mm
Exceeding 750 mm	D + 900 mm	D + 750 mm
All diameters	Not applicable	D + 300 mm

- (d) Notwithstanding the foregoing, any rock in trench excavation shall be so excavated that the clearance between the pipe when laid and the rock sides and bottom of the trench is kept to the minimum limits necessary to provide for the specified thickness of bedding haunching and surround to the pipe. Any excavations outside these limits whether for working space or due to overbreak shall be held to be Excess Excavation.
- (e) The sides of Trench Excavation shall be vertical unless the S.O. permits otherwise. The pipe trench for steel pipelines shall always be vertical from formation level to a minimum of 300 mm above the crown of the pipe.
- (f) Stones shall be removed from the trench invert and holes so formed shall be filled with well compacted selected material. Where mechanical excavators are used, the last 150 mm of the excavation shall be removed by hand to the required depth, and the trench invert carefully trimmed true to level.
- (g) Any widening or deepening of Trench Excavations necessary to accommodate curves, joints or bends in the pipe or to provide extra working space for the construction thereof shall be held to be Excess Excavation.
- (h) Trench Excavation shall wherever practicable be carried out in such a way that every part of the excavation is at least 600 mm clear of the existing edge of the carriageway. In any event the Contractor shall take special precautions, which shall include the continuous support of the sides of the excavation, from the time when excavation begins until the backfilling of the trench is placed, to ensure that there is no disturbance of the adjacent road or road foundation.
- (i) For the purposes of measuring certain work in connection with Trench Excavation "nominal limits" of Trench Excavation are stated above and any excavation outside these limits, which have not been ordered by the S.O. shall be held to be Excess Excavation.

G3.10.12 Excavation in Roads, Road Shoulders and Footpaths

- (a) All excavation and other work carried out within the limits of any road reserve shall be completed as rapidly as possible and not more than half of the width of the carriageway shall be obstructed at one time. A minimum width of 4 m wide of traffic lanes shall be kept clear for flow of traffic and where necessary the shoulder should be strengthened to the satisfaction of relevant Authorities at the Contractor's own cost. Road drains and grids shall be kept free from obstruction.
- (b) Where excavation in public roads, road shoulders or footpaths of congested areas necessitates carting spoil to distant tips, the Contractor shall employ suitable plant to ensure that the passage of traffic past the Works is not impeded by the loading and unloading of spoil. Excavated materials shall not be left standing on road pavement. All trenches on metalled roads shall not be left open for more than one (1) day and all trenches on road verges shall not be left open for more than three (3) days.
- (c) Tarmacadam, concrete and similar road surfaces shall be broken out neatly along the trench line using a diamond disc road cutter to keep the edges straight and vertical.
- (d) The Contractor shall take special precautions, including the continuous support of the sides of the excavation from the time excavation commences until backfilling of the excavation or trench is completed to ensure that there is no disturbance of the adjacent road or road foundation. The Contractor shall also provide barricades to fence off the works and approved road sign boards placed at regular intervals to ensure safety of the public and vehicles to the satisfaction of the S.O. and regulatory authorities.

G3.10.13 Excavation in Fields

- (a) The Contractor shall have regard to the safety of people or livestock in fields or which may be introduced to the fields, and shall ensure that all excavations, access routes and steep or loose slopes arising from the Contractor's operations in these fields are adequately fenced and protected.
- (b) After the erection of temporary fencing the Contractor shall remove topsoil to such depth and over such area as may be necessary to provide sufficient material to ensure adequate surface reinstatement of the working areas occupied by the Contractor for construction of the Works.

G3.10.14 Rock and other Materials in Excavations

- (a) Any material in the excavations which the Contractor considers may be classified as Rock as defined in the general specifications shall be notified to the S.O. before excavation of the material commences. The quantities of Rock excavated from within the nominal limits of excavation shall be recorded by the S.O. and the Contractor each day or at such shorter intervals as the S.O. may require. Only such proportion of material so notified, recorded and signed shall qualify for additional payment as Rock. Overbreak (that is excavation outside the nominal limits of excavation) shall be kept to a minimum and shall be held to be Excess Excavation.

- (b) Materials like shale and weathered sandstone strata may not be classified as 'Rock' within the meaning specified in the general specifications. These materials are tough and may exceed conventional methods of excavation. Bulk excavation of these materials shall include the use of an efficient parallelogram type ripper having a single shank in sound working condition for attachment to a track machine also in sound working condition with a minimum weight of 24 tonnes and a net power rating of not less than 240 hp. These materials shall only be classified as 'Rock' when, in the opinion of the S.O., they cannot be ripped efficiently as described above.

G3.10.15 Supporting Excavation

- (a) The Contractor shall effectively support the sides and ends of all excavations to prevent any fall or run from any portion of the ground outside the excavation and to prevent settlement or damage to structures adjacent to the excavation. Any extra excavation necessary to provide space for such support or other working space shall be held to be Excess Excavation. If, for any reason, any portion of the bottoms, sides or ends of any excavations shall give way, the Contractor shall at his own expense take all necessary remedial measures including the excavation and removal of all the ground thereby disturbed both within and without the nominal limits of excavation and such extra excavation shall be held to be Excess Excavation.
- (b) Where the Contractor elects and is permitted by the S.O. to perform excavations with sloping faces (other than sloping excavations shown on the Drawings or required as permanent features of the Works) and without shoring, the excavated faces shall be to stable slopes and heights and the resulting extra excavation shall be held to be Excess Excavation.

G3.10.16 Trimming Excavations

- (a) When excavating to specified levels for the foundation of any structure or pipes or to specified limits for the faces of any structure required to abut against undisturbed ground, the Contractor shall not excavate the last 150 mm until immediately before commencing the construction work, except where the S.O. shall permit otherwise. Should the Contractor have excavated to within 150 mm above these specified levels or to within 150 mm of these specified limits before he is ready or able to commence the construction work he shall, where required by the S.O., excavate further so as to remove not less than 150 mm of material immediately before commencing the construction work and such further excavation shall be held to be Excess Excavation. Before commencement of any construction work, all shattered and loose material shall be removed from the excavations by hand so as to ensure that the work rests on a solid and perfectly clean foundation or abuts against solid ground.

G3.10.17 Inspection of Excavation

- (a) When the specified levels or limits of excavation are reached, the S.O. will inspect the ground exposed, and if he considers that any part of the ground is by its nature unsuitable he may direct the Contractor to excavate further. Such further excavation shall be backfilled to the specified levels or limits with concrete, selected excavated materials or selected imported material as directed by the S.O. but shall not be held to be Excess Excavation.
- (b) Should the material forming the bottom of any excavation, while acceptable to the S.O. at the time of his inspection, subsequently become unacceptable to him due to exposure to weather conditions or due to flooding or have become puddled, soft or loose during the progress of the Works, the Contractor shall remove such damaged, softened or loosened material and excavate further by hand. Such further excavation shall be held to be Excess Excavation.
- (c) No excavation shall be filled or covered with concrete until it has been inspected and the Contractor has been authorised to proceed with the work. Any concrete or other work put in before the excavations has been inspected and approved by the S.O. shall, if so directed, be removed and new work substituted in accordance with the Specification after the excavations have been approved, all at the Contractor's own expense.

G3.10.18 Temporary Drains

- (a) The whole of the excavation works shall be kept well drained throughout the Contract Period and the Contractor shall ensure that so far as is practicable in the opinion of the S.O. all work is carried out in the dry. The Contractor shall provide temporary drains to deal with all the water which may collect and find its way into the work so that it does not interfere with the work in progress. The temporary drains shall also serve to collect water from the excavated site and prevent excess flow to the adjacent properties.
- (b) The construction of the temporary drains shall be to the approval of the S.O. and the Contractor shall submit to the S.O. full details of the layout and method of drainage he proposes to adopt.
- (c) All cost and charges for dealing with water for all works under the Contract and keeping them dry shall be deemed to be included in the Contract Rates.

G3.10.19 Trenches not to be Left Open

- (a) Unless instructed by the S.O., no length of trench excavation shall be started until the pipes to be laid in that length are available on Site. Trench Excavation shall be carried out expeditiously and, subject to any specific requirements of the Contract, the backfilling and surface reinstatement of Trench Excavations shall commence and be completed as soon as reasonably practicable after the pipes have been laid and jointed. Joint holes may be left open until the joints have been air tested and any external protection has been completed.

- (b) Pipe laying shall follow closely upon the progress of Trench Excavation, and the Contractor shall not permit the carrying out of Trench Excavation in unreasonably excessive lengths to ensure that the pipe laying matches with the progress of Trench Excavation. The Contractor shall take precautions to prevent flotation of pipes in locations where Trench Excavations may be subject to water ingress or flooding and these precautions may include the partial backfilling of the trench leaving pipe joints exposed while awaiting tests of the joints.
- (c) If the S.O. considers that the Contractor is not complying with any of the foregoing requirements he may prohibit further Trench Excavation until he is satisfied with the progress of laying of pipes and backfilling of Trench Excavation.

G3.10.20 Other Structures in Pipelines

- (a) The Contractor shall carry out further excavation as may be necessary to accommodate structures such as thrust blocks and valve chambers. Such excavation shall include for disposal of surplus material and, where appropriate, for backfilling round the structures.

G3.10.21 Temporary Spoil Tips on the Site

- (a) Temporary spoil tips may be used to store excavated material as required and shall be arranged by the Contractor subject to the S.O.'s approval having regard to any particular requirements of the Contract. The location of each temporary spoil tip shall be approved by the S.O. prior to the storing of excavated material at that location.
- (b) Only material which is approved by the S.O. shall be placed in the temporary spoil tips, topsoil being placed in separate spoil tips where so ordered. No tree trunks, stumps, roots, branches or rubbish of any kind shall be placed in spoil tips.
- (c) Temporary spoil tips shall be so shaped as to maintain stability and good drainage at all times.
- (d) The Contractor shall not stockpile excavated materials adjacent to trenches and other excavations unless excavation side slopes and excavation support systems are designed, constructed and maintained for stockpile loads.
- (e) The contractor shall not stockpile excavated materials near or over existing facilities, adjacent property, or completed works, if the weight of stockpiled material could cause damage to existing facilities, adjacent property or completed work.

G3.10.22 Treatment of Material from Excavation

- (a) Subject to any specific requirements of the Contract, the Contractor shall make his own arrangements for the temporary storage of any excavated material which is required for use in backfilling trench excavation, filling to form general platforms and backfilling to general excavation including any necessary double handling. The Contractor shall have regard to the working areas available to him for the construction of the Works. Any temporary tips alongside the excavations shall be to stable slopes and heights and shall be formed so as not to prejudice the stability of trenches, excavations or existing or formed slopes.

- (b) Where the nature of the excavated material is suitable for subsequent incorporation into the permanent works the Contractor's temporary storage as aforesaid shall include for the separate storage as the S.O. may direct of any of the various grades of material specified for the backfilling and surface reinstatement of excavations.
- (c) The Contractor shall ensure that no excavated material which is suitable and is required for reuse in the Works is disposed of off Site.
- (d) Any excavated material not required or not suitable for use in the Works shall be declared surplus and shall become the liability of the Contractor and he shall be entirely responsible for its removal from the Site and for its ultimate disposal. The Contractor shall comply with all the regulations for the disposal of surplus excavated materials off the site including the payment of royalties, fees, etc., which may be imposed by the relevant Authorities.
- (e) Road haulage based methods will be used to remove the surplus excavated material off Site unless otherwise approved by the S.O..
- (f) For materials disposed off Site by truck, the Contractor shall comply with LTA (Roads) regulations regarding excessive loads, covering of trucks, cleaning of wheels and truck bodies, cleaning of roads and adhering to speed limits.
- (g) Slurry shall be retained within site boundary.

G3.10.23 Preparation of Formation for Embankments and Other Fill Areas

- (a) All areas upon which fill is to be placed shall be cleared and stripped of topsoil unless otherwise specified or directed by the S.O.. Excavation shall be carried out to a depth where, in the opinion of the S.O., a suitable formation has been reached. Where fill is to be placed against a formation with a slope greater than 1 in 5, the formation shall be excavated in steps at least 0.5 metres high, so that the fill is keyed into the formation unless otherwise directed by the S.O..
- (b) Except where a drainage blanket is to be placed over soft alluvium and at any other areas as instructed by the S.O., the Contractor shall remove all organic materials and such other materials from the formation of embankments and other fill areas as directed by the S.O.. Where a drainage blanket is to be placed over soft alluvium or other fill material is to be placed over poorly drained ground, bushes and trees shall be cut off at ground level and removed, but roots and other surface vegetation shall be left in place.
- (c) Excavation shall be carried out in the dry. The Contractor shall provide sufficient pumps and equipment to remove the water and maintain a sufficiently low water table to carry out all necessary works.
- (d) The formation of embankments and other fill areas shall be kept free of standing water and foundations not on rock shall be excavated to leave the last 150 mm of material in place until immediately before the placing of fill.
- (e) After excavation and surface preparation has been completed in a particular area the Contractor shall take all necessary steps to prevent deterioration of the surface before filling commences. Notwithstanding these requirements, if any deterioration does take place, the Contractor shall at his own expense carry out such remedial work as the S.O. considers to be reasonably necessary.

- (f) Depressions in the formation of embankments and other fill areas resulting from the removal of unsuitable materials shall be filled and compacted with approved materials as directed by the S.O..
- (g) No fill material shall be placed on any part of the formation until that part has been inspected and approved by the S.O. and agreed levels recorded. Where required by the S.O. the formation shall be sprayed with water before fill material is placed.
- (h) If the method of excavation disturbs the material at or below the surface of the acceptable formation, such disturbed material shall be removed. Any extra excavation and filling necessary because of delays in filling or caused by damage from traffic after the formation has been approved or by disturbance of acceptable formation material by plant or due to any other causes shall be deemed to be Excess Excavation.

G3.10.24 Investigation of Formation for Embankments and Other Fill Areas

- (a) Where directed by the S.O., the Contractor shall carry out additional investigations of the formation for embankments and other fill areas. The Contractor shall excavate exploration pits or sink boreholes not less than 150 mm diameter or carry out in-situ testing of the formation material. Payment for such additional investigation shall be at Contract Rates, or where these are not applicable, at Day work Rates.

G3.10.25 Source of Fill Materials

- (a) Material for use as general, engineering and selected fill shall be obtained from the excavations within the Site unless otherwise specified or authorised in writing by the S.O.. Immediately after material is excavated, it shall be placed to form embankments and general platforms above original or stripped ground levels and compacted as specified without delay. It shall not be deposited in stockpiles unless otherwise directed by the S.O. or allowed to deteriorate by exposure to the weather.

G3.10.26 Placing of Fill and Backfill

- (a) References in this clause to fill and filling shall also be applicable to backfill and backfilling unless specifically stated otherwise.
- (b) Excavated material to be used as fill to form embankments or to raise the ground level or as backfill to excavations and completed structures shall be free from organic material, clods and lumps and shall be approved by the S.O.. Unless otherwise specified, the maximum size of particle used shall be 150 mm. All fill and backfill material shall be well graded. Gap graded materials shall not be used for filling or backfilling.
- (c) Guidance on compaction procedure for earthworks is given in Table G3.10-1 but shall not relieve the Contractor from his obligation to achieve the specified degree of compaction as set out in this clause.

- (d) During compaction, the material for fill shall have a uniform moisture content between the optimum moisture content minus 2 per centum (2%) and the optimum moisture content plus 4 per centum (4%) as recorded in the BS Compaction Test.
- (e) The cost of dewatering or drying including double handling of fill if necessary shall be deemed to be covered by the Contract Rates for fill materials.
- (f) When the weather is such that in the opinion of the S.O. it would adversely affect the placing of fill, all filling and embanking shall be stopped.
- (g) The Contractor shall ensure that a good bond is achieved between layers of fill, and unless otherwise directed by the S.O., no material shall be placed on previously compacted layers until the surface has been harrowed or otherwise broken up and, if necessary, watered. Any unsuitable material shall be removed. The mixing of wet and slurry materials on the embankment shall be prohibited.
- (h) Approval of the proposed surface shall be obtained prior to the Contractor placing fill on a previously compacted area. Any fill on areas not so approved will be liable to be deemed unsuitable by the S.O.. The Contractor shall carry out all necessary remedial work promptly as directed by and to the satisfaction of the S.O..
- (i) Any material not complying with the specified density or moisture content shall be removed from the embankment notwithstanding the fact that it may have been covered by subsequent layers.
- (j) Should the material being placed as fill, while acceptable at the time of selection, become unacceptable to the S.O. due to exposure to weather conditions or due to flooding or have become puddled, soft or segregated during the progress of the Works, the Contractor shall at his own expense remove such damaged, softened or segregated material and replace it with fresh approved material.
- (k) The Contractor shall, when placing the fill or backfill, make due allowance for any settlement that may occur before the end of the Defects Liability Period. Where necessary, the Contractor shall at the end of the Defects Liability Period remove any excess material or make up any deficiency of backfilling or fill to the specified levels.
- (l) Embankments and other fill areas shall be overbuilt by a horizontal distance of not less than 300 mm and shall then be trimmed to the final profile immediately before soiling and grassing.
- (m) Where the Contract requires the placing of different types of fill material in separate layers, the Contractor shall carry out the work to avoid mixing the different types of material. Should there be any mixing of materials, such mixed materials shall be removed from the Site and replaced with separate materials. No additional payment shall be made for such work.
- (n) Drainage materials shall comprise hard durable particles derived from river deposits or crushed rock material and shall be free from silt and clay. Drainage material, after compaction, shall conform to the grading limits specified. Drainage material shall have less than 5% by weight passing a BS 150-micron sieve and such portion shall be non-cohesive.

G3.10.27 Fill Materials for Embankments and Excavations

- (a) General Fill including Backfill:
 - (i) The approved material shall be placed in layers not exceeding 250 mm depth before compaction and shall be compacted by an approved roller to a dry density not less than 95% of the maximum dry density obtained in the BS Compaction Test as described in BS 1377: Part 4; and
 - (ii) Where necessary the Contractor shall adjust the moisture content of the fill either by drying out or adding of water subject to the approval of the S.O.. After such drying out or adding of water the fill shall be thoroughly mixed until the moisture content is suitable.
- (b) Engineering Fill including Backfill:
 - (i) Engineering Fill to form platforms for founding structures and to form embankments shall be used where shown on the Drawings and as directed by the S.O. on Site. Elsewhere General Fill shall be used. Engineering backfill shall be used at locations where components of structures are subsequently to be founded on the backfilled area;
 - (ii) Engineering Backfill shall be used to backfill where material has been excavated beneath the formation level of new or existing structures, where shown on Drawings and as directed by the S.O. on Site;
 - (iii) Engineering fill shall be well graded and the grading curve shall conform to the particle size distribution as specified in the general specifications;
 - (iv) Cohesive and non-cohesive material shall be placed in layers not exceeding 250 mm depth before compaction and shall be compacted by an approved roller to a dry density not less than 97% of the maximum dry density obtained in the BS Compaction Test as described in BS 1377: Part 4. The compacted cohesive material for engineering fill shall additionally have an undrained shear strength between 75 and 110 kPa obtained in the BS Shear Strength Test as described in BS 1377: Part 4; and
 - (v) Where necessary the Contractor shall adjust the moisture content of the fill either by drying out or adding of water subject to the approval of the S.O.. After such drying out or adding of water the fill shall be thoroughly mixed until the moisture content is suitable to obtain the specified dry density for granular materials and dry density and undrained shear strength for cohesive materials.

G3.10.28 Filling of Trench Excavations

- (a) Trench Excavation shall normally be backfilled using suitable materials selected from excavations carried out within 200 m of the length to be backfilled or imported fill materials.

- (b) For trench excavation to be backfilled with excavated material, material free from organic matter and stones of all kinds, shall be deposited in 150 mm layers and thoroughly rammed under and around the pipe with suitably shaped rammers working alternately on either side of the pipe (particular care being taken to avoid damage to the pipe and any sheathing) until the trench has been backfilled up to mid-section of the pipe, and thereafter until the fill has been carried up at least 300 mm above the top of the pipe.
- (c) The remainder of the backfilling may consist of coarse material free from boulders, organic matter and clods of earth larger than 150 mm in size provided that the compacted backfill is, in the opinion of the S.O., sufficiently dense to prevent material from the superimposed layers being washed into the voids in such backfill. This coarse material shall be spread in layers of depth not greater than 225 mm and be thoroughly rammed by an approved mechanical rammer or pan vibrator.
- (d) The coarse filling is to be carried up to the level at which surface reinstatement is to commence or to such level at which the surface reinstatement of the whole of the topsoil, will leave the finished work sufficiently "proud" to allow for future settlement to the original ground level. The reinstated surface shall be rolled with a 2 tonne roller when the material is dried. Any part of the reinstated surface that has settled beyond the adjacent original ground level shall be made good by the Contractor to avoid formation of drains or gulleys within the backfilled trenches.
- (e) For trench excavation to be backfilled with sand material, the sand material shall be deposited in 150 mm layers on both sides of the pipe simultaneously and thoroughly compacted around the pipe working alternately on either side of the pipe until the trench has been filled up to 150 mm above the top of the pipe except for pipes laid in roads where the sand backfill shall be brought up to the base course formation level. The sand filling material shall be compacted by a suitable mechanical vibrator. The topping of the backfilling shall be in accordance with the remainder of the backfilling for trench excavation as described above for backfill with excavated material.
- (f) Unless other special bedding or surround material is specified, backfilling around pipes in trenches excavated in Rock shall be carried out with sand material.
- (g) Sand filling materials such as mining sand or river sand falling within the appropriate grading limits will be accepted for use. Quarry dust shall not be considered for use as filling material.
- (h) Where directed by the S.O. the Contractor shall adjust the moisture content of the backfill material either by drying out or by adding water to assist the compaction of the material.
- (i) Should the material to be placed as backfilling, while acceptable at the time when approved, become unacceptable to the S.O. due to exposure to weather conditions or due to flood or have become puddled, soft or segregated during the progress of the Works, the Contractor shall not use such material for backfilling trenches and shall only use fresh approved material.
- (j) Trench Excavation shall be backfilled with concrete to the height and limit as shown on the Drawings and where directed by the S.O..

- (k) Where the soil is completely unstable, and if, in the opinion of the S.O., large settlements in the pipe line are expected, special arrangements are to be made after proper site investigation and structural calculations. These special arrangements shall be: improvement of mechanical properties of the soil; replacement of soil by other soil or concrete; addition of crushed rock; pile foundation or R.C. slab formation. Where such conditions occur, a report shall be submitted to S.O., or his representative along with all the site data and Contractor's proposal for approval. Where hard rock occurs in the bed just before or after the compressible soft bed, the Contractor shall provide a flexible joint at the junction of two materials to allow rotation of pipe pieces without damage in case of settlement of soft bed. In the case of pipelines laid under the slope of an embankment, any longitudinal tensile or bending stress occurring after the pipe shall be considered.

G3.10.29 Soft Backfill Material - Special Measures

- (a) Where in the opinion of the S.O. sufficient supply of the aforesaid soft material for trench backfilling cannot reasonably be obtained from Trench Excavation within 200 m of the length of trench to be backfilled without resorting to sieving or other special means then the S.O. shall order the Contractor:
 - (i) To carry out such work as may be necessary to sieve out stones, or
 - (ii) To transport suitable soft material from Trench Excavation at distances greater than 200 m from the length to be backfilled ("overhaul"), or
 - (iii) To excavate soft material from suitable, borrow areas and transport it to the length of trench to be backfilled.

G3.10.30 Rockfill Embankments

- (a) Rockfill embankments shall be constructed of approved material from designated or other approved sources. The grading curve of the rockfill shall not be gap graded and shall meet with the approval of the S.O..
- (b) The S.O. reserves the right to make inspections of quarries and pits outside the site. The approval by the S.O. of material from a particular quarry or pit shall not be construed as constituting the approval of all material from that quarry or pit and the Contractor shall be held responsible for the specified quality and gradation of all such materials delivered to site.

G3.10.31 Stone Pitching

- (a) Stone for pitching shall be granite or other clean, hard, dense and durable rock free from cracks, kaolinised patches, organic or other impurities or deleterious materials and shall be approved by the S.O..
- (b) The dimension of each stone measured perpendicular to the face of the pitching shall not be less than 150 mm and shall exceed 225 mm for at least 60% of the pitched area. The exposed face of each stone shall not be less than 155 cm² in area.
- (c) Stone shall be hand placed with closed joints on a layer of gravel or crushed stone to a minimum finished thickness of 225 mm.

- (d) The sides of stones shall be roughly trimmed with a spalling hammer to obtain a reasonably close fit and the interstices filled with clean coarse aggregate or gravel well rammed and wedged with spalls, the finished pitching shall show an even surface to the lines and levels shown on the Drawings.
- (e) The joints in pitching shall be raked out to a depth of at least 20 mm and sealed with 1:3 cement/sand mortar finished neatly flushed with the surrounding stones; the stones being left clean of all mortar stains. Measurement of pitching shall be made by the square metre of surface pitched. The Contract Rates for pitching shall apply whether the surface to be pitched is level or inclined and shall include for all the requirements of this clause.

G3.10.32 Filter Materials for Groundwater Relief Drainage

- (a) Filter material shall consist of single size aggregate conforming to the grading limits specified below. The grading and the materials shall be subject to the approval of the S.O. and they shall not be gap graded. The Contractor shall supply to the S.O. grading curves and samples of not less than 20 kg weight, of each filter material within two weeks after the S.O.'s order to commence work. The S.O. will require to see the curves and samples for each category of material which is in contact with another material in the section of the works concerned.
- (b) The aggregate crushing value of the material shall not exceed 30 per cent. The materials passing the No. 36 BS sieve shall be non-plastic when tested accordance with BS 1377.
- (c) Grading of Filter Material B: The grading curve of the Filter Material B shall conform to the following particle size distribution and shall meet with the approval of the S.O.:

<u>BS Sieve</u>	<u>Percentage passing by weight</u>
200 mm	80-100
60 mm	30-80
20.0 mm	0-30
10.0 mm	0-12
3.35 mm	0

- (d) Grading of Filter Material C: The grading curve of the Filter Material C shall conform to the following particle size distribution and shall meet with the approval of the S.O.:

<u>BS Sieve</u>	<u>Percentage passing by weight</u>
37.5 mm	100
9.2 mm	45-100
3.35 mm	25-80
600 µm	8-45
75 µm	0-10

- (e) When filter material is used with perforated pipes at least 15 per cent of the material shall be larger in particle size than twice the maximum dimension of the perforations.
- (f) Filter material shall be compacted in layers no exceeding 150 mm to the optimum moisture content pre-determined by laboratory testing. The contractor shall be deemed to have allowed in his Contract Rates for placing filter material for any watering necessary to achieve the required moisture content.

G3.10.33 uPVC Perforated Pipe for Groundwater Relief Drainage

- (a) Where shown on the Drawings, uPVC perforated pipes shall be used for ground water relief drainage. UPVC perforated pipe shall be made from rigid PVC and shall conform to BS 3506. Pipes shall be 150 mm nominal diameter, longitudinally grooved on the exterior and have a smooth internal surface with an integrally moulded socket. Internal surface shall have the operating roughness $k = 0.7$ mm.
- (b) The uPVC perforated pipe shall have a tunnel-shaped cross-section with slot size width of 0.8 mm and radially slotted, 4 offset rows spread over 220 degrees of the pipe crown.
- (c) The water shall be disposed of lawfully without damage to new and existing facilities or adjoining properties.

G3.10.34 Composite Filter Membrane with Drainage Core

- (a) Where shown on the Drawings a composite filter membrane with drainage core should be fixed against the walls of structures and laid under base slabs. The composite shall consist of a black polyethylene core grid to collect and channel water into drainage pipes and with polypropylene geotextile on both side to keep out fine soil particles from blocking the drainage mat.
- (b) The composite material consisting of core grid and filter layer shall weigh 1000 g/m² and shall have a nominal thickness of between 5.0 and 10.0 mm thick. The hydraulic properties of the composite material shall comply with the following table for an effective pore size of filter material O90W = 0.12 mm:

δ (kPA)	2	10	20	200	400
kV (m/s)	4.2E-3	3.4E-3	2.8E-3	2.3E-3	1.9E-3
kH (m/s)	2.5E-1	1.9E-1	1.4E-1	1.2E-1	8.2E-2
q (l/s.m)	0.465	0.370	0.348	0.302	0.270

- (c) The geotextile used to sandwich the core grid shall have a minimum punch resistance of 3.6kN and 39% for punch distortion when tested in accordance with DIN 54307. In addition, the filter material should be able to withstand a maximum tensile force of 16kN/m in both longitudinal and transverse direction and not exceeding 25% and 33% in tensile elongation in the longitudinal and transverse respectively.
- (d) The composite material shall be fixed using bitumen mastic applied directly onto a clean concrete wall face in 100 mm wide bands along the edges of the areas covered and by laying down the mat directly onto the layer of compacted under blanket prior to base slab construction.

G3.10.35 Placing of Fill - Groundwater Relief Drainage Construction

- (a) Areas upon which filter material for groundwater relief drainage are to be placed shall be prepared to form a sensibly smooth formation free of vertical projections. The Contractor shall not place the filter material until the formation surfaces have been prepared and approved as specified.
- (b) The groundwater relief drainage construction shall be laid and fixed in the following sequences:
 - (i) Beneath Base of Structures
 - 1) A layer of under blanket material of 350 mm thick Type B filter material shall be laid as shown on the Drawings and compacted in two equal layers prior to the laying of the composite filter membrane above the drainage under blanket layer;
 - 2) When laying the composite filter membrane on top of the drainage under blanket, the Contractor should ensure that no fines or sharp vertical protrusions are visible which hinder the performance of the drainage system. Heavy gauge polyethylene sheeting shall then be laid on top of the composite filter membrane immediately prior to placing concrete blinding; and
 - 3) The composite membrane material shall be adequately overlapped or butted at the interfaces between horizontal and vertical drainage planes with a layer of similar geotextile sealing the joint in accordance with the manufacturer's recommendations and to the satisfaction of the S.O. to ensure effective continuity of the groundwater relief drainage system.
 - (ii) Structural Walls
 - 1) The composite filter membrane material shall be securely fastened to the concrete face as specified to the satisfaction of the S.O.;
 - 2) A vertical layer of filter material Type C shall then be placed immediately adjacent to the composite filter membrane material to form a vertical trench drain as shown on the Drawings. The Type C material shall be compacted in layers of 250 mm. The compaction of filter material shall be carried out as specified and the Contractor shall maintain the required moisture content by watering if necessary; and
 - 3) 150mm diameter uPVC perforated drainage pipes shall be installed to collect the groundwater adjacent to structures at levels and locations as shown on the Drawings or as directed by the S.O. on Site and backfilling continued as specified in paragraph (e.) above.
- (c) The Contractor should ensure that his rates allows for trimming, overlapping of the composite and excavating and backfilling or the use of temporary formwork to separate different materials or any other operation necessary to comply with these requirements.

G3.10.36 Care of Existing Services

- (a) Information relating to the positions of all existing mains, cables, culverts, pipes, drains or services of any kind shall be ascertained by the Contractor. Notwithstanding any information which may be furnished by the Board or the S.O., the Contractor shall be responsible for ascertaining from his own inspection of the Site and from the respective supply authorities and other public authorities the positions of all mains, pipes and cables whether underground or overhead, within or near the Site. Where services are shown on the Drawings or where there is evidence of services in the ground, the Contractor shall also determine their exact positions. In addition, the Contractor shall watch for and determine the position of any service which may not have been marked.
- (b) Where excavation is carried out close to or across the line of sewers, pipes, cables or other services the Contractor shall provide temporary adequate supports to secure the services and where such sewer, pipe, cable or other service is temporarily displaced or disturbed it shall be restored to its original state. If restoration of any service to its original state cannot be achieved then it shall be replaced and made good. The cost of excavating around any service which crosses the excavation, and of temporarily supporting it while construction is undertaken, shall be deemed to be included in the rates for excavation.
- (c) All damage to services shall be made good to the satisfaction of the S.O. and the Authorities concerned. The cost of making good any damage shall be deemed to be included in the rates for excavation. If any damaged service shall only be repaired by the Authority concerned, all costs whatsoever relating to the repair of the damaged service undertaken by the authority shall be at the Contractor's expense. Where damage to a service had already existed prior to excavation or other work, repair of the damaged service shall be undertaken by the Contractor, if so ordered by the S.O. and the Contractor shall be paid on Day work for repairing such service.
- (d) Where in the opinion of the S.O., excavation and/or subsequent construction cannot reasonably be carried out unless the sewer, pipe, cable or other service is permanently severed or permanently diverted or permanently supported the Contractor shall undertake such work if so ordered by the S.O.. Any such work if carried out shall be to the satisfaction of the S.O. and of the Authority concerned and shall be paid for a Day work where necessary and at the appropriate Contract Rates.
- (e) Where pipes are laid under culverts, they shall subject to other provisions in the Contract and to the S.O.'s instructions, be provided with a concrete surround extending 600 mm clear on each side of the culvert, and a concrete half haunch to the culvert shall be provided over the width of the trench cast directly on top of the concrete surround. The cost of all work in removing any piles and concrete bed under culverts and of temporarily supporting the culvert during construction shall be deemed to be covered by the Contract Rates.

G3.10.37 Land Drains

- (a) Where land drains, mole drains or field drains are severed by excavation they shall be kept in effective temporary operation during construction. The Contractor shall submit to the S.O. for approval a temporary drainage arrangement to ensure the watercourse is not hindered during construction.
- (b) The drain on either side of the Trench Excavation shall be cut back for at least 300 mm and temporary pipes of suitable length and diameter shall be jointed to the existing drain and laid resting with the ends on solid ground with suitable stopping to prevent the subsequent run of land drainage water into the pipe trench. During trench backfilling, the backfill material shall be carefully placed and thoroughly compacted under the temporary pipes to give them adequate support. At the appropriate stage of backfilling the Trench Excavation the drains shall be permanently restored to the satisfaction of the S.O.. Should any existing sub soil or field drains be uncovered during general excavation, the Contractor shall either carefully replace them when backfilling, or if this is impracticable shall divert them to new drains or ditches, or otherwise relay them as the S.O. may direct.

G3.10.38 Hedges, Fences and Walls

- (a) Where construction passes under barriers such as hedges, fences and walls the Contractor shall, as a temporary measure, provide temporary fencing for any parts of such barriers as have had to be removed.
- (b) After excavations have been backfilled and surfaces reinstated, the Contractor shall carry out work as the S.O. may order for permanent restoration of such barriers. In the case of a hedge, the section removed shall be replaced by new plants of the appropriate variety and where ordered by the S.O. the plants shall be protected from livestock on both sides by an adequate post and barbed wire fence. During the Defects Liability Period, all hedges replanted in the above manner shall be inspected and any dead plants shall be replaced by the Contractor.

G3.10.39 Surface Reinstatement in Road Verges, Fields, etc.

- (a) Where excavation is in road verges, fields and grassed areas, the turf previously removed may be used for surface reinstatement provided that the grass has not been killed during stacking and that the surface is free from all loose stones.
- (b) After filling of the excavation as specified, the Contractor shall replace all topsoil previously removed and it shall be evenly distributed and levelled over the full extent of the stripped area. The working space occupied by the Contractor which was originally covered with grass, if destroyed during construction, shall be covered with 75 mm of topsoil, close turfed and maintained until the new grass is properly established at the Contractor's expense. Other areas not originally covered with grass shall be restored to their original state.
- (c) Road verges shall be restored to their original width and to a minimum fall of 1 in 24 away from the road and towards the drainage ditch (if any). Restoration of the cross fall and drainage ditches shall be carried out unless the fact that no cross fall or ditch existed prior to pipe laying has been recorded in writing and agreed to by the S.O..

- (d) The Contractor shall restore all surfaces to a condition not inferior to that which existed prior to commencement of the Works.
- (e) Any damage to metalled roads due to Work alongside road verges or shoulders shall be reinstated to the satisfaction of the Authority concerned at no extra cost to the Board.

G3.10.40 Surface Reinstatement in Roads and Footpaths

- (a) Surface reinstatement of roads and footpaths shall be restored to the approval of the S.O. and to the requirements of the Authorities concerned or the owner or tenant. Any settlement or defects prior to final surface reinstatement shall be made good by the Contractor to the satisfaction of the S.O..
- (b) Where Trench Excavation crosses or is along tarmacadam roads and footpaths belonging to private owners or the Authorities, the pipe shall be surrounded with sand and the trench backfilled with crusher run material followed by 100 mm thick of dense bituminous mix comprising 60 mm thick binder course and 40 mm thick wearing course. The crusher run and binder course surfaces shall receive a layer of tack coat before the application of binder course material and wearing course materials. All fill materials shall be compacted by approved means.
- (c) The consolidated surface shall be brought up to the same level as the adjoining road. Any subsidence shall be filled in by the Contractor using the same materials. The road shall be kept continuously at its proper level to the satisfaction of the S.O. and the Authorities from the date of filling of the trench until the surface is finally restored. All materials for this reinstatement and subsequent maintenance shall be supplied by the Contractor.
- (d) When any section has been reinstated, the Contractor shall notify the S.O. who in the presence of the Contractor shall inspect the length of reinstatement. The S.O. shall at this inspection indicate any further reinstatement required and the Contractor shall complete this work to the satisfaction of the S.O. and the Authorities within fourteen (14) days of the inspection. Acceptance of reinstatement by the S.O. or the Authorities at this inspection shall not absolve the Contractor of his responsibility for the maintenance of surfaces as specified above.

G3.10.41 Works in Roads and Road Reserves

- (a) Where Work is carried out within the road reserve, the Contractor shall comply fully with all regulations and requirements in force in that place by all relevant local and public authorities referred to in the Specification elsewhere as Authorities.
- (b) Working hours within road reserves will be subject to the conditions imposed by the Authorities. The working hours may or may not fall within day light hours or within the normal working hours of the Authorities.
- (c) The Authorities may impose charges for Works carried out on road verges and metalled roads and the Contractor shall pay the Authorities and include the cost in his Contract Rates.

G3.10.42 Contractor's Supervision of Embanking, Filling and Backfilling

- (a) The Contractor shall provide continuous supervision of excavation, filling and backfilling by an experienced staff of engineers and foremen. No excavation, filling or backfilling shall be carried out unless and until the Contractor provides such supervision.
- (b) The Contractor shall be responsible for carrying out all field and laboratory tests required for the proper control of the earthworks. Laboratory tests shall be carried out in a specially designed laboratory established in accordance with the general specifications.
- (c) A daily record, in English, shall be kept of materials placed in the embankments, fill or backfill and materials run to spoil areas including spoil areas reserved for condemned material and two copies shall be handed to the S.O. daily.

G3.10.43 Taking and Testing Undisturbed Samples for Independent Verification

- (a) The S.O. may require the Contractor to take and arrange to have tested undisturbed samples in an approved independent laboratory. Where directed by the S.O., the Contractor shall take undisturbed samples from anywhere on the Site and in the Works and shall deliver the samples to such approved independent laboratories as may be appropriate for the tests required. The Contractor shall ensure that testing is carried out expeditiously and shall furnish to the S.O. two (2) copies of the results of such tests. A provisional item has been included in the Bill of Quantities to cover all costs associated with the taking and independent testing of undisturbed samples.
- (b) Undisturbed cylindrical samples 100 mm in diameter and 450 mm long and undisturbed block samples 250 mm x 250 mm x 250 mm shall be taken from the formation and compacted fill where directed by the S.O. and shall be protected until they are tested.
- (c) The samples shall be taken and protected as described in BS EN ISO 22475-1 "Geotechnical Investigation and Testing – Sampling Methods and Groundwater Measurements".

G3.10.44 Testing by the Contractor

- (a) The Contractor shall be responsible for carrying out all field and laboratory tests to ensure that all fill material placed complies with the grading limits specified, undrained shear strength (where applicable) and plasticity index (where specified), that the material is compacted to within the specified limits of dry density and at moisture contents within the limits specified (where applicable). Testing shall be to BS 1377 unless otherwise approved.
- (b) Particle size distribution (grading) shall be determined by the appropriate test from BS 1377: Part 2. The tests described in BS 1377: Part 2 shall be used unless otherwise agreed by the S.O..
- (c) Fill density shall be determined by replacement methods as follows:
 - (i) In fine, medium and coarse-grained soils, sand replacement shall be employed generally in accordance with BS 1377: Part 9.; and

- (ii) In rockfill and reno mattress, where directed by the S.O., water replacement shall be employed using a polythene lining film in a test hole not less than 0.4 m³ in volume.
- (d) Samples for laboratory testing of maximum dry densities and optimum moisture contents shall be on samples taken adjacent to and including in-situ density test locations.
- (e) Maximum dry densities and optimum moisture contents for compaction control by relative compaction shall be derived using the test described in BS 1377: Part 4 and maximum and minimum dry densities for compaction control by relative density shall be derived using the test described in BS 1377: Part 4.
- (f) The location of each field test or sampling point shall be to the approval of the S.O.. The minimum frequency of control testing required per 1,000 m³ of fill placed, or per day where the volume of fill placed on that day is less than 1,000 m³, shall be as set out in Table G3.10-2.
- (g) In addition to routine soil testing carried out at the frequency given in Table G3.10-2, the S.O. may instruct additional tests to be carried out to verify that the fill materials placed conform to the Specification.
- (h) The S.O. shall be given 24 hours' notice prior to the commencement of testing and shall be given ready access to the laboratory at all times. Test results and copies of calculations shall be submitted promptly upon completion to the S.O.. The S.O. shall be solely responsible for interpreting the results of the tests.

G3.10.45 Testing by the S.O.

- (a) The carrying out of any tests by or on behalf of the S.O. shall not relieve the Contractor of any of his responsibilities for testing in accordance with the Specification.

G3.10.46 Anti-Termite Treatment

- (a) Approved anti-termite treatment shall be carried out through specialist who possess acceptable experience in this field and who can provide references of successful guaranteed work having been carried out earlier. The solution used for anti-termite treatment shall be an approved solution applied strictly as per manufacturer's instructions and shall be capable of offering a minimum of 10 years warranty against termite attack. Equal alternative chemical solutions and the instructions on how to apply anti-termite treatment can be proposed by the Contractor for the S.O.'s approval.

G3.10.47 Top Soiling and Turfing

- (a) Topsoil for grassed areas and planting shall be fertile natural red/brown topsoil obtained from within the Site unless the S.O. permits otherwise. Topsoil shall be free from large stones, roots, sticks, clay, peat, weeds and sods, and shall be obtained from naturally well-drained areas. Topsoil shall be neither excessively acid nor alkaline and shall be free of any toxic matter liable to be harmful to plant growth.

- (b) Earthworks slopes shall be trimmed and cultivated as soon as possible after each cutting or embankment has been completed. The Contractor shall establish grass by continuous close turving as directed.
- (c) All areas to be turfed shall be covered with topsoil taken from stockpile approved by the S.O. and lightly rolled or tamped.
- (d) Topsoil shall be evenly spread and trimmed over embankments and other areas to the slopes and levels shown on the Drawings or ordered by the S.O.. The depth after spreading and trimming shall be 100 mm unless otherwise directed, measured normal to the surface. All clods and lumps shall be broken up and any rubbish, stones, roots and weeds shall be removed.
- (e) The prepared areas shall then be closed turfed with turf laid to a well bonded pattern with no gaps between the turfs and lightly beaten. The turf grass shall be *Axonopus Compressus* (common name "Cow or Field" grass).
- (f) All turf used for cultivation shall be not less than 250 mm square and shall be obtained from an approved source and shall be formed of dense, well established grass free from weeds or other plant growth and shall not contain stones, sticks or other undesirable material. The thickness of the turf root mat shall be not less than 40 mm.
- (g) All turf shall be cut and lifted only immediately prior to delivery of the Site and, shall be kept moist during transit. Delivery of turf to the Site shall be so scheduled as to allow final placing within 24 hours of arrival on Site.
- (h) Where turf is laid on slopes steeper than 1:3, each turf shall be securely pegged down with two cleft wooden pegs 150 mm long.
- (i) All turfed areas shall be lightly rolled with an approved roller or punned immediately after laying and shall be watered regularly until the grass has been established.
- (j) The Contractor shall be responsible until the end of the Defects Liability Period for the establishment of the turf growth in the cultivated areas and for the replacement of topsoil eroded from the prepared areas. In periods of dry weather, the Contractor shall water the cultivated areas as necessary to sustain growth. The Contractor shall, as and when instructed by the S.O. during the construction of the Works or during the Defects Liability Period re-cultivate at his own expense any areas:
 - (i) In which the turf has died;
 - (ii) In which, in the opinion of the S.O., the turf has not established sound growth within four months of being laid; or
 - (iii) Which have become eroded of turf or topsoil.
- (k) During the period of the Works and the Defects Liability Period the Contractor shall keep all cultivated areas mown or cut at a frequency at least once a month to the satisfaction of the S.O..

G3.10.48 Payment for Trench Excavation

- (a) Payment for trench excavation shall be made in two stages as follows:
 - (i) At 70% of the Contract Rates after completion of pipe laying and backfilling excluding final reinstatement; and
 - (ii) The remaining 30% of the Contract Rates after completion of pipe laying, restoration of ground surfaces including reinstatement of all surfaces damaged during pipe laying work, completion of chambers, thrust blocks, concrete surrounds, etc.
- (b) Payment for the second stage shall be due only after restoration of any length has been inspected by the S.O. and thereafter certified as satisfactory by the S.O..

G3.10.49 Payment for Turfing

- (a) Payment for continuous turfing shall be for the area covered by the turf provided. Where spot turfing is required the distance between the centre lines of divots in the form of a triangular pattern shall not exceed 500 mm and the percentage of the area covered shall be agreed and payment made on the total area spot turfed multiplied by the percentage.
- (b) Turfing will be paid for in two parts:
 - (i) At 50% of the Contract Rate when the turfing has been carried out; and
 - (ii) The remaining 50% of the Contract Rates when the S.O. is satisfied that the grass has established itself.
- (c) Where grass fails to establish itself, the Contractor shall turf the relevant area again at no extra cost to the Board.

G3.10.50 Testing

- (a) Except where items have been provided specifically in the Bills of Quantities for the testing of earthworks, the cost of such testing, including obtaining samples, their handling and transportation and the provision of test results all as specified, shall be deemed to be included in the Contract Rates for earthworks.

G3.10.51 Notes Relating to Table G3.10-1

- (a) The notes given in this clause and the table below are for guidance only and shall not relieve the Contractor from his basic obligation to produce work to the required standard of the Specification.
 - (i) The depth of a compacted layer is the height by which an embankment or ground level is raised by each successive compacted layer.
 - (ii) Number of passes is the number of times that each point on the surface of the layer being compacted has been traversed by the item of compaction plant. In assessing the number of passes of pneumatic tyred rollers to meet the requirements of Table G3.10-1, the effective width shall be the sum of the widths of the individual wheel tracks together with the sum of spacing between the wheel tracks provided each spacing does not exceed 225 mm for Table G3.10-1. When the spacing exceeds the above, the effective width shall be the sum of the widths of the individual wheel tracks.

- (iii) The force per 100 mm width is the total force (weight) on the roll divided by the total roll width. Where a smooth wheeled roller has more than one axle the machine will be assessed based on the axle giving the highest value of force per 100 mm width.
- (iv) For pneumatic tyred rollers, wheel load is the total weight of the roller divided by the number of wheels.
- (v) Vibratory rollers are self-propelled or towed having means of applying mechanical vibration to one or more rolls:
 - 1) The requirement for vibratory rollers are based on the use of the lowest gear on a self-propelled machine and towing speed of 1.5 - 2.5 km per hour for a towed machine. If higher gears or speeds are used an increased number of passes shall be provided in proportion to the increase in speed of travel;
 - 2) Vibratory rollers operating without their vibration mechanism in use will be classified as smooth wheeled rollers;
 - 3) Vibratory rollers shall be operated with their vibration mechanism operating only at the frequency of vibration recommended by the manufacturers. All such rollers shall be equipped with a device automatically indicating the frequency at which the mechanism is operating; and
 - 4) Combination of different types of plant will be permitted in which case the number of passes for each type shall follow such proportion of the appropriate number as will together produce the same total compactive effort as any type operated singly.
- (vi) Vibrating plate compactors are machines having a base plate to which is attached a source of vibration consisting of one or two eccentrically weighted shafts:
 - 1) The static pressure under the plate of a vibrating plate compactor is calculated by dividing the total weight of the machine in working order by the area in contact with compacted soil; and
 - 2) Vibrating plate compactors shall be operated at the frequency of vibration recommended by the manufacturers. They shall normally be operated at travelling speeds of less than 15 m/min but if higher speeds are necessary the number of passes shall be increased in proportion to the increase in speed of travel.
- (vii) Vibro tampers are machines in which an engine driven reciprocating mechanism acts on a spring system, through which oscillations are set up in a base plate.
- (viii) Power rammers are machines which are actuated by explosions in an internal combustion cylinder, each explosion being controlled manually by the operator.
- (ix) In the case of power rammers one pass will be considered as made when the compacting shoe has made one strike on the area in question.

- (x) For items marked with an asterisk the rollers shall be towed by track laying tractors. Self-propelled rollers are unsuitable.
- (xi) Where combinations of different types or categories of plant are used, the compaction requirements shall be:
 - 1) the depth of layer shall be that for the type of plant requiring the least depth of layer; and
 - 2) the number of passes shall be that for the type of plant requiring the greatest number of passes.
- (xii) However, where the Contractor uses a lighter type of plant to provide some preliminary compaction only these requirements to assess the effective compaction could be disregarded.

**TABLE G3.10-1: COMPACTION PROCEDURE FOR EARTHWORKS AND SUBGRADES
FOR GUIDANCE ONLY**

Type of Compaction on Plant	Category	Cohesive Soils		Well-graded Granular & Dry Cohesive Soils		Uniformly-graded Materials	
		Maximum depth of compacted layer (mm)	Minimum number of passes	Maximum depth of compacted layer (mm)	Minimum number of passes	Maximum depth of compacted layer (mm)	Minimum number of passes
Smooth-wheeled roller	Force per 100mm width (kN)						
	2.1-2.6 2.61-5.2 > 5.2	125 125 150	8 6 4	125 125 150	10 8 8	125 125 Unsuitable	10* 8* Unsuitable
Grid-roller	Force per 100mm width (kN)						
	2.6-5.2 5.3-7.8 > 7.8	150 150 150	10 8 4	Unsuitable 125 150	Unsuitable 12 12	150 Unsuitable Unsuitable	10 Unsuitable Unsuitable
Vibratory roller	Force per 100mm width (kN)						
	0.25-0.45 0.46-0.70 0.71-1.25 1.26-1.75 1.76-2.30 2.31-2.80 2.81-3.50 3.51-4.20 4.21-4.90	Unsuitable Unsuitable 100 125 150 175 200 225 250	Unsuitable Unsuitable 12 8 4 4 4 4 4 4	75 75 125 150 150 175 200 225 250	16 12 12 8 4 4 4 4 4	150 150 150 200 225 250 (+) (+) (+) (+)	16 12 6 10* 12* 10* (+) (+) (+) (+)
Pneumatic-tyred roller	Wheel Load (t)						
	1.0-1.5 1.5-2.0 2.0-2.5 2.5-4.0 4.0-6.0 6.0-8.0 8.0-12 > 12	125 150 175 225 (+) (+) (+) (+)	6 5 4 4 (+) (+) (+) (+)	Unsuitable Unsuitable 125 125 125 150 150 175	Unsuitable Unsuitable 12 10 10 8 8 6	150 Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable	10* Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable Unsuitable

* These rollers shall be towed. Self-propelled rollers are unsuitable

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**TABLE G3.10-1: COMPACTION PROCEDURE FOR EARTHWORKS AND SUBGRADES (CONT'D)
 FOR GUIDANCE ONLY**

Type of Compaction on Plant	Category	Cohesive Soils		Well-graded Granular & Dry Cohesive Soils		Uniformly-graded Materials	
		Maximum depth of compacted layer (mm)	Minimum number of passes	Maximum depth of compacted layer (mm)	Minimum number of passes	Maximum depth of compacted layer (mm)	Minimum number of passes
Vibrating- plate compactor	Static pressure under base plate: (kN/m ²)						
	8.6-10.3 10.3-12.1 12.1-13.8 13.8-17.2 17.2-20.7 > 20.7	Unsuitable Unsuitable Unsuitable 100 150 200	Unsuitable Unsuitable Unsuitable 6 6 6	Unsuitable 75 75 125 150 200	Unsuitable 10 6 6 5 5	75 100 150 150 200 250	6 6 6 4 4 4
Vibro- tamper	Mass (kg)						
	50-65 65-75 > 75	100 150 200	3 3 3	100 125 150	3 3 3	150 200 225	3 3 3
Power rammer	Mass (kg)						
	100 > 500	150 250	4 8	150 250	6 12	Unsuitable Unsuitable	Unsuitable Unsuitable

* These rollers shall be towed. Self-propelled rollers are unsuitable.

(+) Unsuitable for layers less than 250 mm thick

TABLE G3.10-2: TABLE OF FREQUENCY OF TESTING OF FILL BY THE CONTRACTOR

Material	A	B	C	D	E	F
General Fill (cohesive soils)	1	2	2	1	1	-
General Fill (non-cohesive soils)	1	2	2	1	-	-
Engineering Fill (cohesive soils)	2	4	4	3	2	4
Engineering Fill (non-cohesive soils)	2	4	4	3	-	-
Sand Fill	3	2	-	-	-	-
Rockfill / Reno Mattress	2	-	-	-	-	-
Test Reference						
A - Particle size distribution or grading						
B - In situ, dry density by sand replacement method						
C - In situ moisture content						
D - Maximum dry density and optimum moisture content (BS 1377)						
E - Plasticity index						
F - Undrained shear strength						

1. The location of each field test or sampling point shall be to the approval of the S.O..

Table G3.10.2 sets out the minimum frequency of control testing required per 1,000 m³ of fill placed, or per day where the volume of fill placed on that day is less than 1,000 m³.

For the purposes of this table a cohesive soil is defined as a soil from which a plasticity index (plastic limit) can be obtained from the component of the soil passing a 425 µm sieve.

END OF SECTION